LAB 19

Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	Monitoring and reporting
TSC reference	A8, A10
Duration	25 min
Simulator page	http://localhost:8000/#/analytics
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

After a year of paper trading, the investment committee wants a one-page tearsheet and a short report on the automated EMA 10/30 strategy: how the research backtest compares with the forward test, what it cost, whether any monitoring limit is breached, and whether the desk should allocate capital - produced automatically by the pipeline, every time.

Objective

Produce a one-page tearsheet and explain the gap between backtest and forward test.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 19. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 19: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Export the tearsheet

Backtest page (or Analytics) > PDF tearsheet for your best backtest; also export the equity and trades CSV.

Open: http://localhost:8000/#/analytics (Analytics > PDF tearsheet)

Check: The PDF has equity vs benchmark, drawdown, monthly heatmap, metrics and the cost table.

Step 2. Backtest vs forward

Analytics: select your Lab 16 bot under 'Bot (backtest vs forward)'.

Open: http://localhost:8000/#/analytics

Check: Explain every material gap (fills, sizing, costs, timing).

Step 3. Draw insights

Monthly heatmap: in which regimes did the strategy earn its returns, what drove its drawdowns?

Open: http://localhost:8000/#/analytics

Check: Two or three insights, not just numbers.

Expected in the app: A PDF tearsheet and a written explanation of the backtest-vs-forward differences.

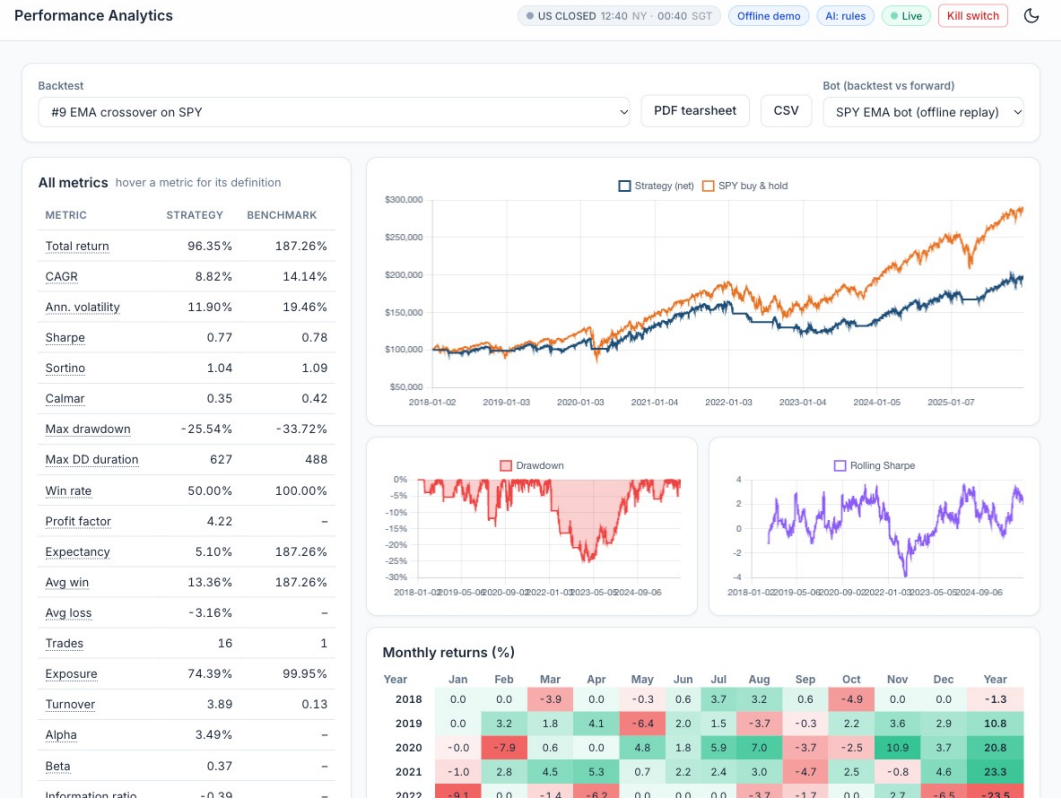


Figure: Lab 19 in the Trading Simulator.

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Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.

A one-page performance tearsheet - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: produce a one-page PDF tearsheet for an EMA 10/30 crossover on SPY 2018-2025 (trade at the next day's open with a 0.05% cost per trade): equity curve against buy-and-hold, a drawdown chart, a monthly returns heatmap and a metrics table (CAGR, Sharpe, Sortino, maximum drawdown, win rate). Then replay 2025 day by day with the same rule and explain any difference from the 2025 backtest.

Open in Claude | **Open in ChatGPT** | **Open Google Colab**

Debrief

Comparing the backtest with the forward test is the final test of whether actions and outcomes match the model (A8). Automated monitoring with explicit thresholds turns that comparison into an alarm, and a good report turns numbers into trends, insights and a clear, conditional recommendation (A10) - including the honest message that the strategy reduces risk but has not beaten buy and hold.

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